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Simulation-Based Inference for an Adaptive-Network Epidemic Model

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GitHub:

https://github.com/KyNoxed/Baptiste_ST3247_Assignment

Based on the project specification by Alexandre Thiéry.

Contents

1 Introduction

1.1 The Model

We consider a SIR epidemic model (Susceptible, Infected, Recovered) on an adaptive contact network. The population consists of $N = 200$ agents, and recovered agents are permanently immune.

The network is initialised as an Erdős-Rényi random graph $G(N, p)$ with $p = 0.05$, giving an average of 10 contacts per agent. Five agents are randomly infected at time $t = 0$. At each discrete time step, three mechanisms operate, governed by three unknown parameters:

- **Infection** (β): each susceptible neighbour of an infected agent becomes infected with probability β.
- **Recovery** (γ): each infected agent recovers with probability γ and moves to the Recovered state.
- **Rewiring** (ρ): for each edge between a susceptible and an infected agent, the susceptible can sever the link and reconnect to a randomly chosen non-infected agent with probability ρ, modelling behavioural avoidance.

This rewiring mechanism makes the model adaptive: the network co-evolves with the epidemic, creating a feedback loop between contact structure and disease dynamics. Simulations run for $T = 200$ time steps.

1.2 Observed Data

We have $R = 40$ independent realisations of the model generated with the same unknown parameters (β, γ, ρ). For each realisation, three observables are recorded:

- the **fraction of infected individuals** over time (`infected_timeseries.csv`),
- the **number of rewiring events** at each time step (`rewiring_timeseries.csv`),
- the **final degree histogram**, i.e. the distribution of the number of contacts per agent at the end of the simulation (`final_degree_histograms.csv`).

The contact network itself is never directly observed.

1.3 Inference Problem and Intractability of the Likelihood

The goal is to estimate (β, γ, ρ) from the data. Classical approaches would maximise the likelihood or build a posterior via Bayes' rule. Here, computing the exact likelihood would require summing over all possible network trajectories — up to $2^{\binom{200}{2}}$ graphs — which is entirely infeasible.

We therefore use Approximate Bayesian Computation (ABC): since we cannot evaluate the likelihood, we instead simulate data from the model for candidate parameter values and compare the simulated output to the observations using summary statistics. Parameters that produce a close match are accepted as approximate posterior samples.

2 Basic Rejection ABC

2.1 Algorithm

Rejection ABC is the simplest simulation-based inference method. The algorithm proceeds as follows:

1. Draw candidate parameters (β, γ, ρ) from the prior.
2. Simulate the model to generate a synthetic dataset.
3. Compute summary statistics and compare them to the observed summaries using a distance function.
4. Accept the parameters if the distance is below a threshold ε; otherwise discard them.

The accepted parameters form an approximation of the posterior $p(\beta, \gamma, \rho \mid \text{data})$.

2.2 Summary Statistics

Each dataset is compressed into a vector of 8 summary statistics:

- **Infected time series:** peak infection fraction, timing of the peak (normalised by total duration), mean infected fraction, and final infected fraction.
- **Rewiring time series:** average rewiring rate and timing of peak rewiring activity.
- **Degree distribution:** mean and standard deviation of the final degree distribution.

The infected time series primarily informs β and γ, while the rewiring and degree statistics provide additional signal about ρ.

2.3 Distance Function and Tolerance

We use a normalised mean squared distance between simulated and observed summaries:

$$d(s_{\text{sim}}, s_{\text{obs}}) = \frac{1}{8} \sum_{k=1}^{8} \left(\frac{s_{\text{sim}}^{(k)} - s_{\text{obs}}^{(k)}}{|s_{\text{obs}}^{(k)}| + \epsilon_0} \right)^2 \tag{1}$$

where $\epsilon_0 = 10^{-9}$ avoids division by zero. Normalising by the observed value ensures all statistics contribute on a comparable scale.

The threshold ε is set to the 1st percentile of all distances over $N = 100{,}000$ simulations, retaining the 1,000 closest parameter draws. This data-driven choice avoids fixing an arbitrary threshold.

2.4 Results

Figure 1 shows the marginal posterior distributions. The posterior for β concentrates around 0.17, γ around 0.09, and ρ around 0.30.

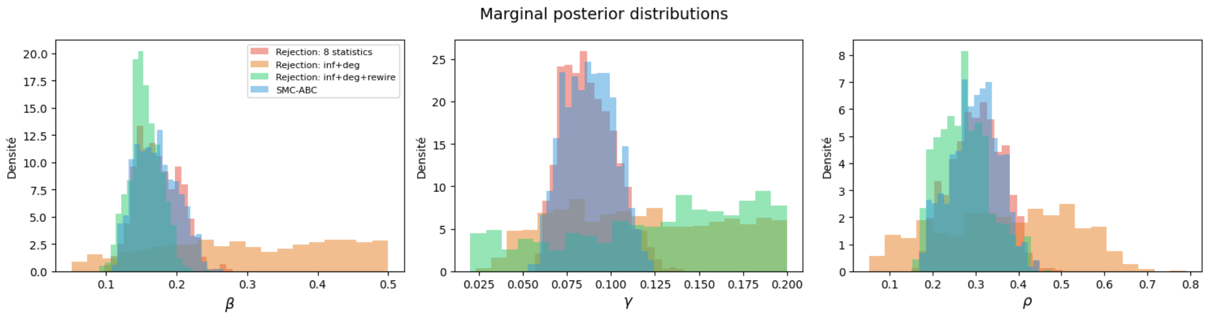

Figure 1: Marginal posterior distributions for β, γ, and ρ obtained by rejection ABC with $N = 100{,}000$ simulations and a 1% acceptance threshold.

2.5 Discussion

Rejection ABC provides a useful first approximation but has clear limitations. The 1% acceptance rate means 99% of simulations are discarded, which is computationally wasteful. Furthermore, β and ρ can both suppress the epidemic through different mechanisms, making them potentially hard to separate from infection curves alone. These limitations motivate the investigations in Sections 3 and 4.

3 Summary Statistics Design

3.1 Motivation

The choice of summary statistics is critical in ABC: poorly chosen statistics yield wide, uninformative posteriors regardless of the number of simulations. A key challenge here is that β and ρ can both suppress the epidemic through different mechanisms — fast transmission vs. early contact-breaking — so statistics based solely on the infected fraction may fail to distinguish them. Rewiring counts and the final degree distribution provide complementary signal: rewiring activity is directly driven by ρ, and the degree distribution reflects how much the network was restructured.

3.2 Compared Configurations

We compared three configurations, each evaluated with $N = 10{,}000$ simulations and a 1% acceptance threshold:

- **Configuration A** – full time series: mean squared error on the infected fraction, rewiring, and degree histogram directly (`dist_inf + dist_deg + dist_rewire`).
- **Configuration B** – partial time series: same as A but excluding rewiring (`dist_inf + dist_deg`).
- **Configuration C** – scalar summaries: the 8 statistics described in Section 2. These were chosen by domain intuition; their effectiveness is validated empirically by the posterior predictive check distance.

3.3 Results

Table 1 reports posterior means and posterior predictive check (PPC) distances. PPC distances are computed using each configuration's own distance function and are therefore not directly comparable in absolute terms, but serve as a relative goodness-of-fit indicator within each configuration.

Table 1: Comparison of summary statistic configurations (N = 10,000, 1% acceptance threshold).

Configuration	$\hat{\beta}$	$\hat{\gamma}$	$\hat{\rho}$	PPC mean	PPC median
A: `inf + deg + rewire`	0.155	0.126	0.274	1.521	1.582
B: `inf + deg`	0.300	0.118	0.369	0.094	0.096
C: scalar summaries	0.174	0.088	0.304	0.015	0.016

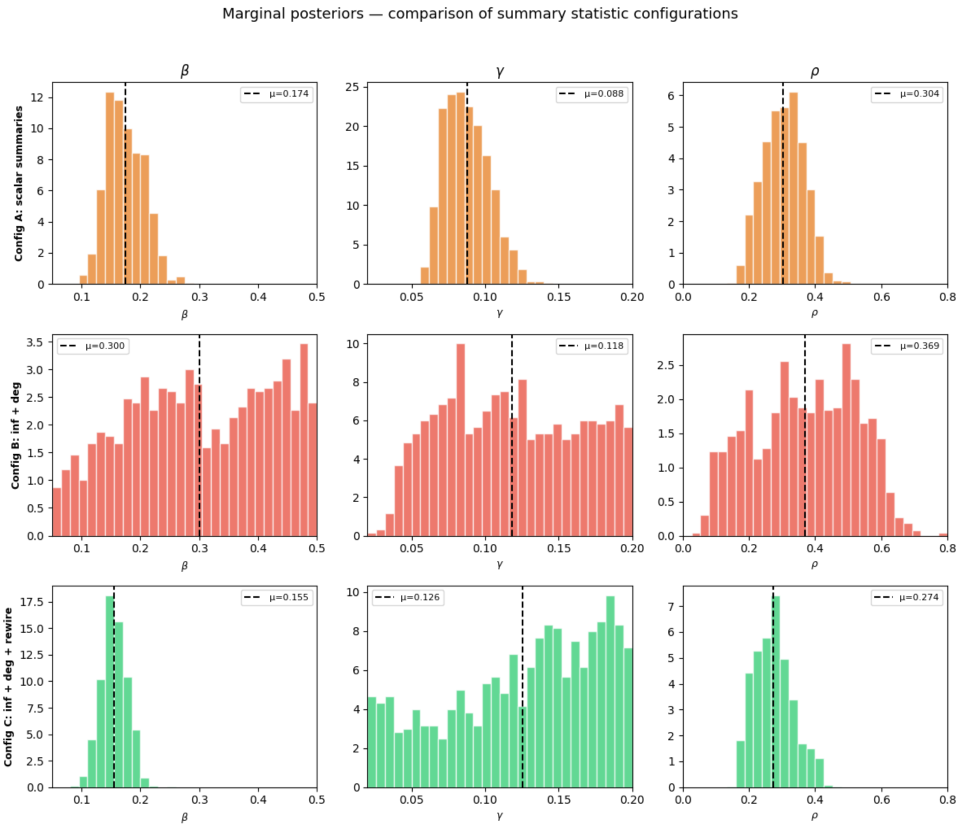

Figure 2: Marginal posteriors for β, γ, and ρ under the three configurations, evaluated over the full prior range. Dashed lines indicate posterior means. Configuration B produces nearly flat posteriors across all three parameters, indicating that β, γ, and ρ are largely unidentified without rewiring information. Configurations A and C yield more concentrated posteriors for β, with C achieving the tightest concentration. γ remains uncertain across all configurations except A. For ρ, Configurations A and C both yield concentrated posteriors.

3.4 Discussion

Raw time series distances hurt rather than help. Configuration A yields the worst PPC distance (1.521 vs. 0.015 for C). The likely cause is a scaling issue: long time series dominate the total distance and overshadow other components, causing the acceptance step to select parameters that fit the distance metric but not the actual data.

Omitting rewiring shifts $\hat{\rho}$ significantly. Configuration B gives $\hat{\rho} = 0.37$, noticeably higher than A and C ($\hat{\rho} \approx 0.27$–0.30), and produces nearly flat posteriors for both β and ρ. This confirms that without rewiring-specific statistics, β and ρ are difficult to disentangle.

Scalar summaries achieve the best fit. Configuration C obtains the lowest PPC distance and the most concentrated posterior for β, as visible in Figure 2. The 8 hand-crafted scalars strike a good balance between informativeness and dimensionality, motivating their use throughout the rest of this report.

4 Advanced Methods: SMC-ABC

4.1 Method

Sequential Monte Carlo ABC (SMC-ABC) replaces the single threshold ε of rejection ABC with a decreasing sequence $\varepsilon_1 > \varepsilon_2 > \cdots > \varepsilon_T$, iteratively refining a population of particles towards regions of high posterior probability. Each iteration proceeds as follows:

1. **Resampling**: draw a particle θ^* from the current population with probability proportional to its weight.
2. **Perturbation**: perturb θ^* with a Gaussian kernel of bandwidth $2\hat{\Sigma}$, where $\hat{\Sigma}$ is the weighted covariance of the current population.
3. **Simulation and acceptance**: simulate the model and accept the candidate if its distance to the observed data is below ε_t.
4. **Importance weighting**: reweight accepted particles to correct for the perturbation bias:
$$w_i \propto \frac{\pi(\theta_i)}{\sum_j w_j^{(t-1)}\, K(\theta_i \mid \theta_j^{(t-1)})} \tag{2}$$
   where π is the (uniform) prior and K is the perturbation kernel.

Since particles are guided towards promising regions from the first iteration, SMC-ABC wastes far fewer simulations than rejection ABC.

4.2 Experimental Setup

We ran SMC-ABC with $N = 1{,}000$ particles and threshold sequence:

$$\varepsilon \in \{0.5,\ 0.3,\ 0.15,\ 0.08,\ 0.04,\ 0.02\} \tag{3}$$

The scalar summaries from Configuration C were used throughout, and the initial population was obtained by rejection ABC at $\varepsilon_1 = 0.5$.

4.3 Results

Table 2 reports posterior means, standard deviations, and acceptance rates at each step. Figure 3 shows the evolution of the marginal posteriors.

Table 2: SMC-ABC posterior estimates across iterations.

Step	ε	Acc. rate	$\hat{\beta}$	$\hat{\gamma}$	$\hat{\rho}$
0	0.50	–	–	–	–
1	0.30	0.234	0.307 ± 0.117	0.119 ± 0.046	0.363 ± 0.215
2	0.15	0.190	0.299 ± 0.115	0.123 ± 0.042	0.387 ± 0.174
3	0.08	0.137	0.253 ± 0.095	0.109 ± 0.035	0.368 ± 0.136
4	0.04	0.061	0.190 ± 0.044	0.096 ± 0.024	0.320 ± 0.086
5	0.02	0.071	0.172 ± 0.030	0.088 ± 0.022	0.300 ± 0.071

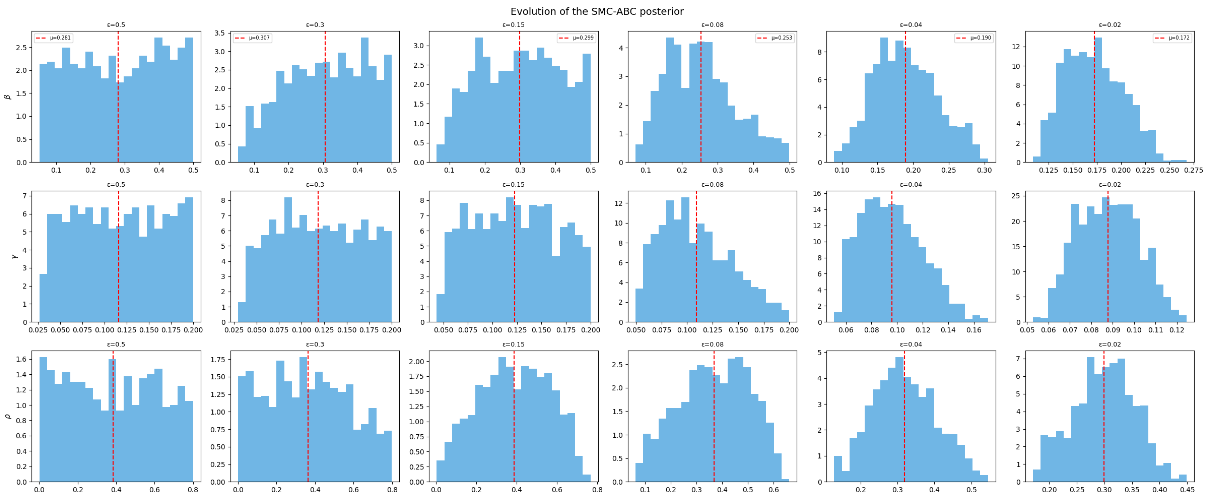

Figure 3: Evolution of the marginal posteriors for β, γ, and ρ across SMC-ABC iterations. Each column corresponds to one threshold ε, decreasing left to right. Dashed red lines indicate weighted posterior means.

4.4 Discussion

Convergence. Acceptance rates remain healthy throughout (6.1%–23.4%), indicating a well-calibrated threshold sequence. Posterior means stabilise noticeably between steps 4 and 5, suggesting convergence.

Progressive concentration. Standard deviations decrease monotonically across all parameters, most clearly for ρ ($\pm 0.215 \to \pm 0.071$) and β ($\pm 0.117 \to \pm 0.030$), confirming that each iteration successfully discards the least compatible particles.

Final estimates and comparison with rejection ABC. The final estimates $\hat{\beta} = 0.172$, $\hat{\gamma} = 0.088$, $\hat{\rho} = 0.300$ are in close agreement with rejection ABC under Configuration C ($\hat{\beta} = 0.174$, $\hat{\gamma} = 0.088$, $\hat{\rho} = 0.304$). This cross-method consistency strengthens confidence in the estimates. The key advantage of SMC-ABC is not a different answer, but a more efficient and reliable path: rather than a single blind pass over the prior, it produces a monitored sequence of increasingly refined posteriors.

5 Conclusion

This report investigated simulation-based inference for an adaptive-network epidemic model with an intractable likelihood. We estimated (β, γ, ρ) using rejection ABC and SMC-ABC, with particular attention to the design of summary statistics.

Three findings stand out. First, the choice of summary statistics is decisive: raw time series distances (Configuration A) and omitting rewiring (Configuration B) both yield unreliable posteriors, while 8 scalar summaries (Configuration C) achieve the best predictive fit. Second, β and ρ are difficult to identify without rewiring-specific statistics, confirming that these parameters can produce similar epidemic curves through distinct mechanisms. Third, SMC-ABC improves upon rejection ABC not by changing the final estimates, but by reaching them more efficiently with a built-in convergence diagnostic.

Our best estimates consistently point to $\hat{\beta} \approx 0.17$, $\hat{\gamma} \approx 0.09$, and $\hat{\rho} \approx 0.30$. A natural extension would be a formal sensitivity analysis of the summary statistics, or the use of neural posterior estimation (NPE) for further gains in sample efficiency.